SUBSCRIPTION MANAGEMENT SYSTEM

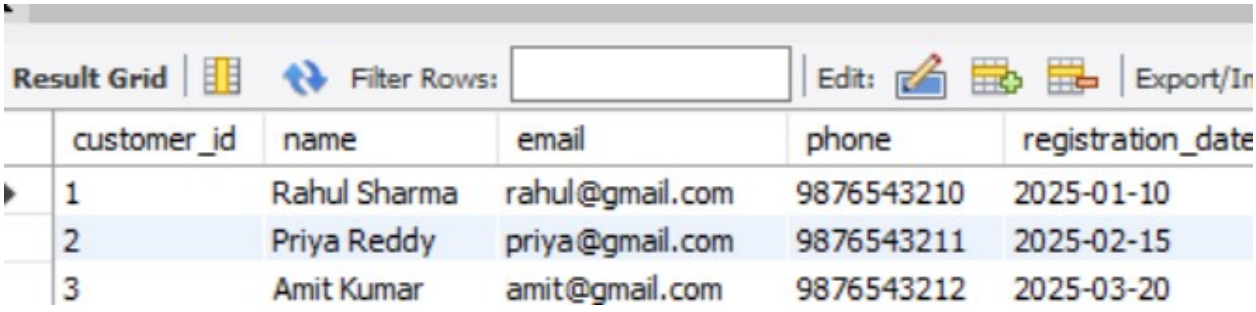
TABLES

1.CUSTOMER TABLE:

```
CREATE TABLE Customers (
    customer_id INT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE,
    phone VARCHAR(15),
    registration_date DATE
);
```

INSERT:

```
INSERT INTO Customers VALUES
(1,'Rahul Sharma','rahul@gmail.com','9876543210','2025-01-10'),
(2,'Priya Reddy','priya@gmail.com','9876543211','2025-02-15'),
(3,'Amit Kumar','amit@gmail.com','9876543212','2025-03-20'),
(4,'Sneha Rao','sneha@gmail.com','9876543213','2024-11-05'),
(5,'Kiran Patel','kiran@gmail.com','9876543214','2025-04-01');
```



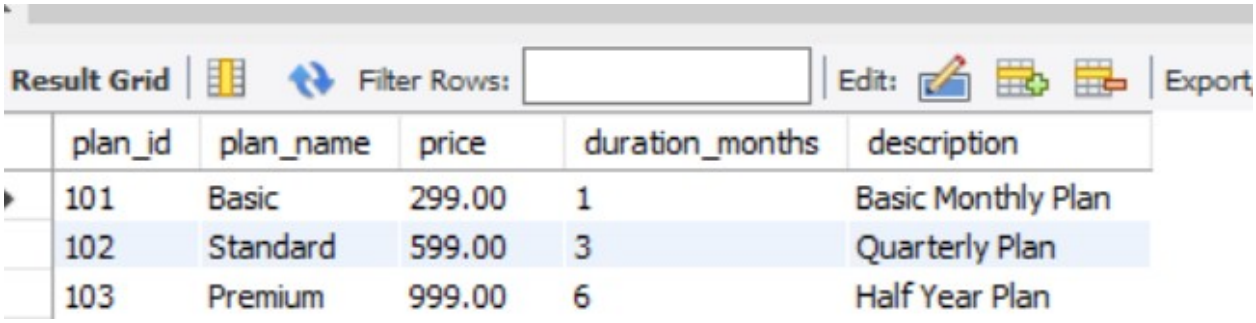
customer_id	name	email	phone	registration_date
1	Rahul Sharma	rahul@gmail.com	9876543210	2025-01-10
2	Priya Reddy	priya@gmail.com	9876543211	2025-02-15
3	Amit Kumar	amit@gmail.com	9876543212	2025-03-20

2.PLANS TABLE:

```
CREATE TABLE Plans (
    plan_id INT PRIMARY KEY,
    plan_name VARCHAR(100),
    price DECIMAL(10,2),
    duration_months INT,
    description VARCHAR(255)
);
```

INSERT:

```
INSERT INTO Plans VALUES
(101,'Basic',299.00,1,'Basic Monthly Plan'),
(102,'Standard',599.00,3,'Quarterly Plan'),
(103,'Premium',999.00,6,'Half Year Plan'),
(104,'Gold',1499.00,12,'Annual Gold Plan'),
(105,'Student',199.00,1,'Student Offer');
```



plan_id	plan_name	price	duration_months	description
101	Basic	299.00	1	Basic Monthly Plan
102	Standard	599.00	3	Quarterly Plan
103	Premium	999.00	6	Half Year Plan

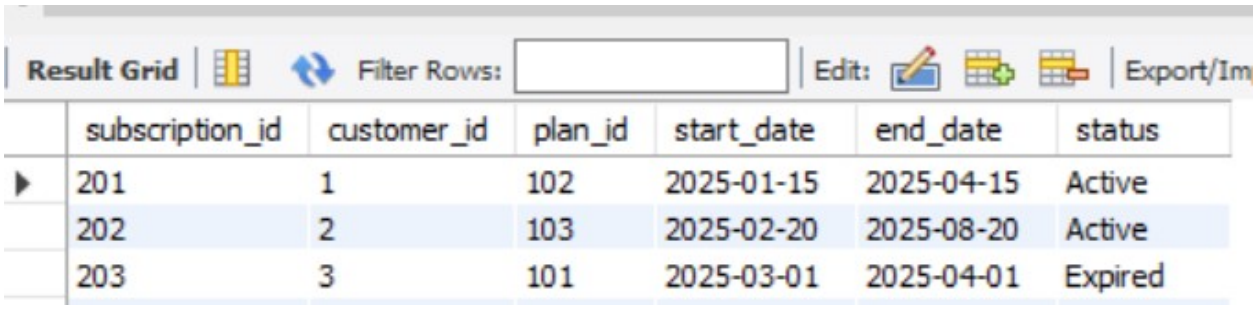
3.SUBSCRIPTION TABLE:

```
CREATE TABLE Subscriptions (
    subscription_id INT PRIMARY KEY,
    customer_id INT,
    plan_id INT,
    start_date DATE,
```

```
    end_date DATE,
    status VARCHAR(50),
    FOREIGN KEY (customer_id) REFERENCES Customers(customer_id),
    FOREIGN KEY (plan_id) REFERENCES Plans(plan_id)
);
```

INSERT:

```
INSERT INTO Subscriptions VALUES
(201,1,102,'2025-01-15','2025-04-15','Active'),
(202,2,103,'2025-02-20','2025-08-20','Active'),
(203,3,101,'2025-03-01','2025-04-01','Expired'),
(204,4,104,'2024-11-10','2025-11-10','Active'),
(205,5,105,'2025-04-05','2025-05-05','Cancelled');
```



subscription_id	customer_id	plan_id	start_date	end_date	status
201	1	102	2025-01-15	2025-04-15	Active
202	2	103	2025-02-20	2025-08-20	Active
203	3	101	2025-03-01	2025-04-01	Expired

4.PAYMENT TABLE:

```
CREATE TABLE Payments (
    payment_id INT PRIMARY KEY,
    subscription_id INT,
    amount DECIMAL(10,2),
    payment_date DATE,
    payment_method VARCHAR(50),
    status VARCHAR(50),
    FOREIGN KEY (subscription_id) REFERENCES Subscriptions(subscription_id)
```

);

INSERT:

INSERT INTO Payments VALUES

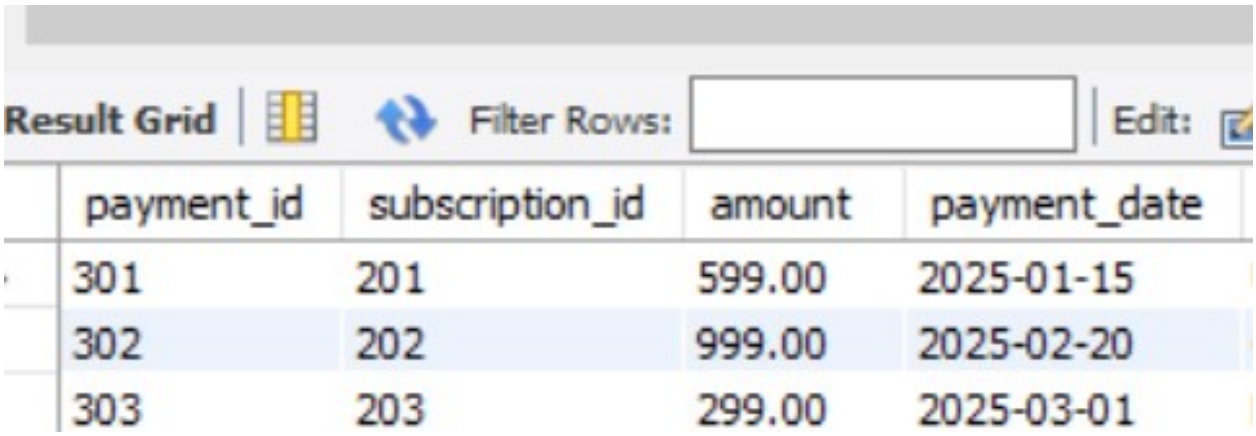
(301,201,599.00,'2025-01-15','UPI','Success'),

(302,202,999.00,'2025-02-20','Credit Card','Success'),

(303,203,299.00,'2025-03-01','Debit Card','Success'),

(304,204,1499.00,'2024-11-10','Net Banking','Success'),

(305,205,199.00,'2025-04-05','UPI','Failed');



payment_id	subscription_id	amount	payment_date
301	201	599.00	2025-01-15
302	202	999.00	2025-02-20
303	203	299.00	2025-03-01

5.USAGE_DETAILS TABLE:

CREATE TABLE Usage_Details (

usage_id INT PRIMARY KEY,

subscription_id INT,

usage_date DATE,

usage_amount INT,

FOREIGN KEY (subscription_id) REFERENCES Subscriptions(subscription_id)

);

INSERT:

INSERT INTO Usage_Details VALUES

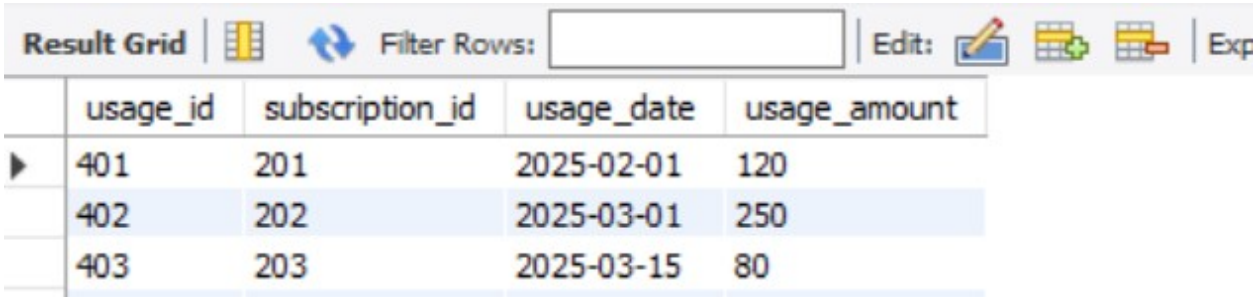
(401,201,'2025-02-01',120),

(402,202,'2025-03-01',250),

(403,203,'2025-03-15',80),

(404,204,'2025-01-20',500),

(405,205,'2025-04-10',50);



usage_id	subscription_id	usage_date	usage_amount
401	201	2025-02-01	120
402	202	2025-03-01	250
403	203	2025-03-15	80

6.DISCOUNT TABLES

CREATE TABLE Discounts (

discount_id INT PRIMARY KEY,

plan_id INT,

discount_percentage DECIMAL(5,2),

valid_from DATE,

valid_to DATE,

FOREIGN KEY (plan_id) REFERENCES Plans(plan_id)

);

INSERT:

INSERT INTO Discounts VALUES

(501,101,10.00,'2025-01-01','2025-12-31'),

(502,102,15.00,'2025-01-01','2025-12-31'),

(503,103,20.00,'2025-01-01','2025-12-31'),

(504,104,25.00,'2025-01-01','2025-12-31'),

(505,105,30.00,'2025-01-01','2025-12-31');

Result Grid | Filter Rows: | Edit: | Export

discount_id	plan_id	discount_percentage	valid_from	valid_to
501	101	10.00	2025-01-01	2025-12-31
502	102	15.00	2025-01-01	2025-12-31
503	103	20.00	2025-01-01	2025-12-31

BASIC QUERIES:

- **DISPLAY PLANS WITH PRICE GREATER THAN 500**

```
SELECT *

FROM plans

where price>500;
```

Result Grid | Filter Rows:

plan_id	plan_name	price	duration_month
102	Standard	599.00	3
103	Premium	999.00	6
104	Gold	1499.00	12

- **SHOW ALL ACTIVE SUBSCRIPTIONS**

```
SELECT *

FROM subscriptions

where status='Active';
```

Result Grid | Filter Rows: | Edit: | Export/I

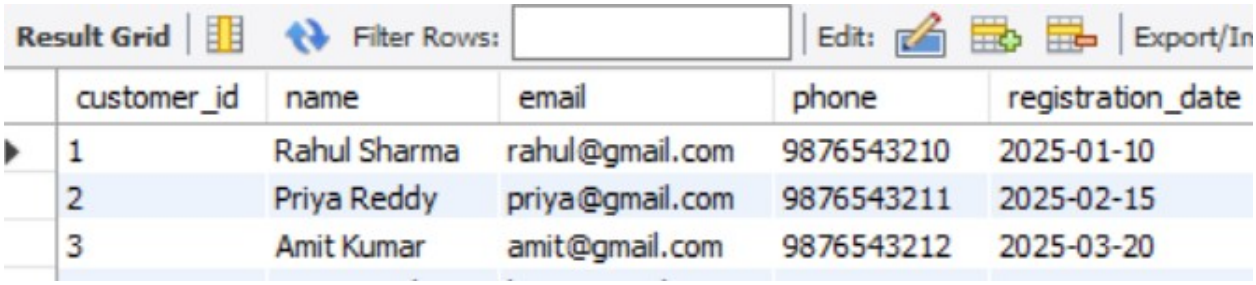
subscription_id	customer_id	plan_id	start_date	end_date	status
201	1	102	2025-01-15	2025-04-15	Active
202	2	103	2025-02-20	2025-08-20	Active

- **LIST CUSTOMERS REGISTERED AFTER 2024**

SELECT *

FROM CUSTOMERS

WHERE REGISTRATION_DATE>'2024-12-31';

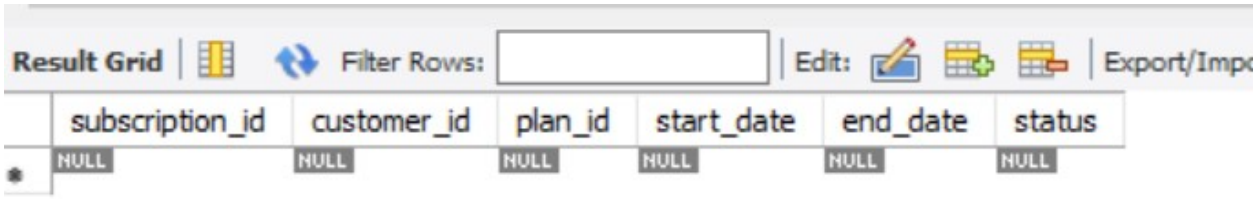


Result Grid | Filter Rows: | Edit: | Export/In

customer_id	name	email	phone	registration_date
1	Rahul Sharma	rahul@gmail.com	9876543210	2025-01-10
2	Priya Reddy	priya@gmail.com	9876543211	2025-02-15
3	Amit Kumar	amit@gmail.com	9876543212	2025-03-20

- DISPLAY SUBCRIPTIONS ENDING THIS MONTH

SELECT * FROM SUBSCRIPTIONS WHERE MONTH(END_DATE)=MONTH(CURDATE());



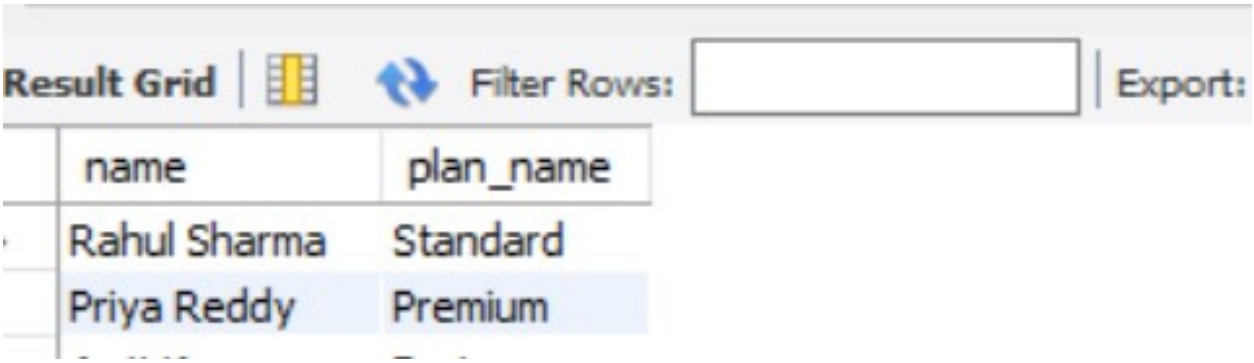
Result Grid | Filter Rows: | Edit: | Export/Impo

subscription_id	customer_id	plan_id	start_date	end_date	status
NULL	NULL	NULL	NULL	NULL	NULL

JOINS:

- **SHOW CUSTOMER NAMES WITH THEIR SUBSCRIPTION PLANS**

SELECT c.name,p.plan_name from customers c join subscriptions s on c.customer_id = s.customer_id join plans p on s.plan_id = p.plan_id;

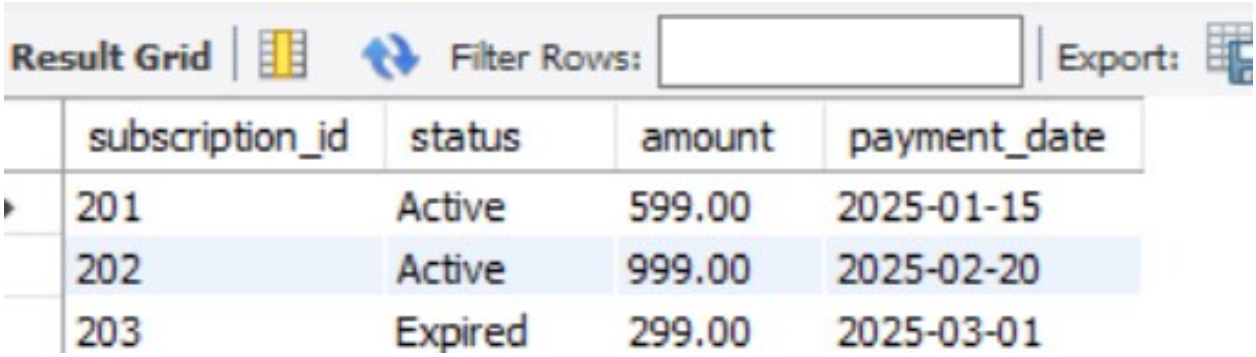


Result Grid | Filter Rows: | Export:

name	plan_name
Rahul Sharma	Standard
Priya Reddy	Premium

- **DISPLAY SUBSCRIPTION DETAILS WITH PAYMENT INFORMATION**

SELECT s.subscription_id, s.status,p.amount,p.payment_date from subscriptions s join payments p on s.subscription_id = p.subscription_id;

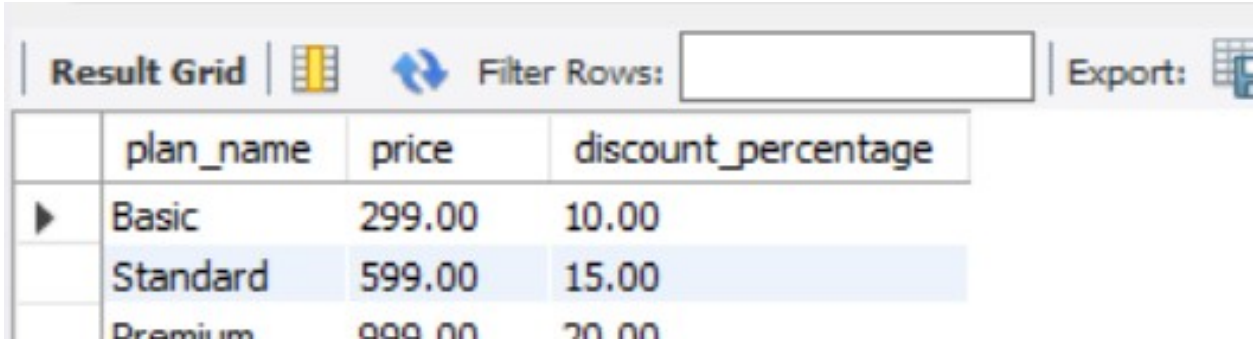


Result Grid | Filter Rows: | Export:

subscription_id	status	amount	payment_date
201	Active	599.00	2025-01-15
202	Active	999.00	2025-02-20
203	Expired	299.00	2025-03-01

- **SHOW PLANS WITH DISCOUNT DETAILS**

SELECT p.plan_name,p.price,d.discount_percentage from plans p join discounts d on p.plan_id = d.plan_id;

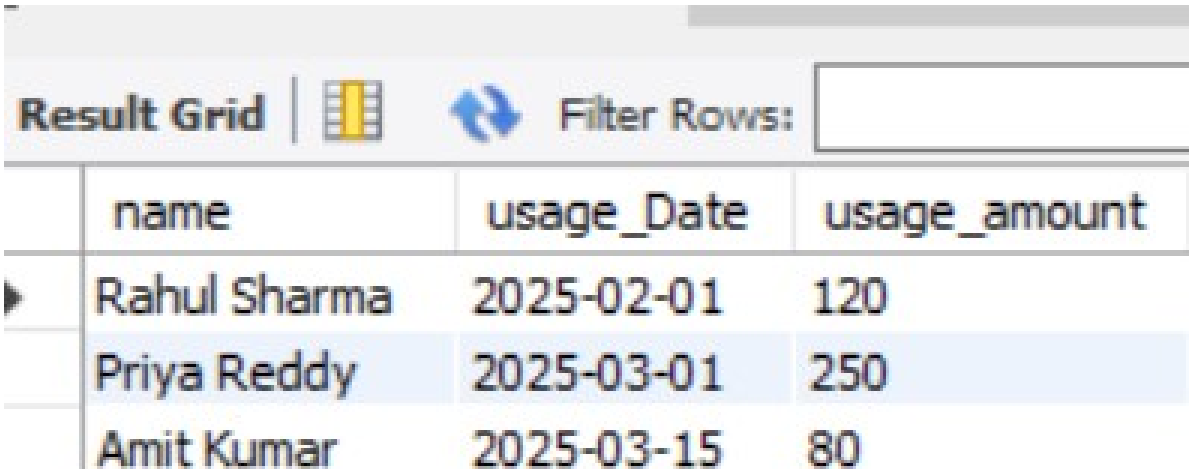


Result Grid | Filter Rows: | Export:

plan_name	price	discount_percentage
Basic	299.00	10.00
Standard	599.00	15.00
Premium	999.00	20.00

- **LIST USAGE DETAILS WITH CUSTOMER NAMES**

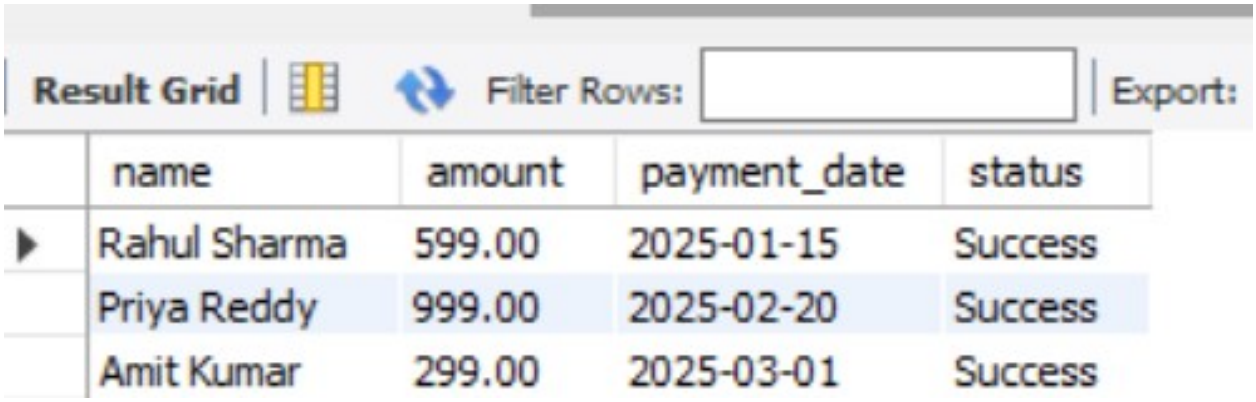
SELECT c.name,u.usage_Date,u.usage_amount from customers c join subscriptions s on c.customer_id=s.customer_id join usage_details u on s.subscription_id = u.subscription_id;



name	usage_Date	usage_amount
Rahul Sharma	2025-02-01	120
Priya Reddy	2025-03-01	250
Amit Kumar	2025-03-15	80

- **SHOW CUSTOMERS WITH THEIR PAYMENT HISTORY**

SELECT c.name,p.amount,p.payment_date,p.status from customers c join subscriptions s on c.customer_id = s.customer_id join payments p on s.subscription_id=p.subscription_id;



name	amount	payment_date	status
Rahul Sharma	599.00	2025-01-15	Success
Priya Reddy	999.00	2025-02-20	Success
Amit Kumar	299.00	2025-03-01	Success

SUBQUERIES

- **FIND CUSTOMERS WITH NO SUBSCRIPTIONS**

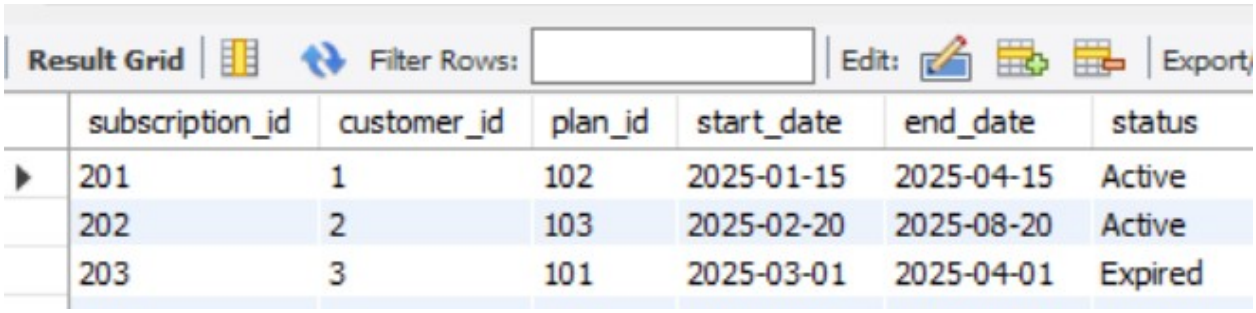
SELECT * FROM CUSTOMERS WHERE CUSTOMER_ID NOT IN(SELECT CUSTOMER_ID FROM SUBSCRIPTIONS);



customer_id	name	email	phone	registration_date
NULL	NULL	NULL	NULL	NULL

- **FIND EXPIRED SUBSCRIPTIONS**

SELECT * FROM SUBSCRIPTIONS WHERE END_DATE<CURDATE();



Result Grid | Filter Rows: | Edit: | Export/

subscription_id	customer_id	plan_id	start_date	end_date	status
201	1	102	2025-01-15	2025-04-15	Active
202	2	103	2025-02-20	2025-08-20	Active
203	3	101	2025-03-01	2025-04-01	Expired

- **SHOW PLANS WITH PRICE GREATER THAN AVERAGE**

SELECT * FROM PLANS WHERE PRICE>(SELECT AVG(PRICE) FROM PLANS);

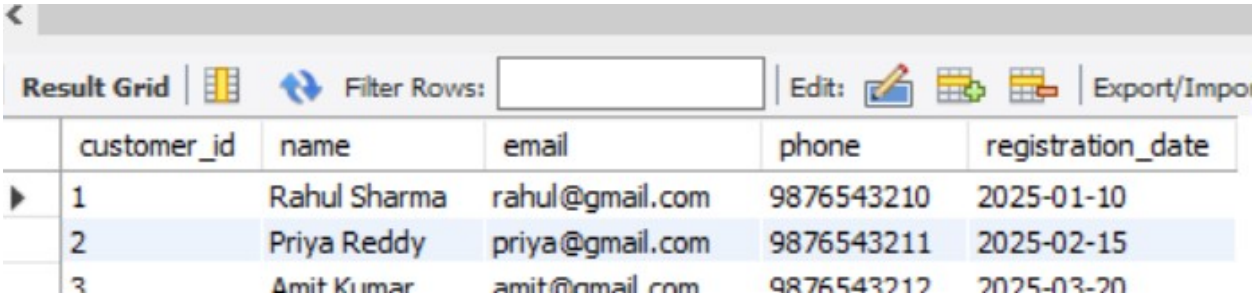


Result Grid | Filter Rows: | Edit:

plan_id	plan_name	price	duration_months	de
103	Premium	999.00	6	Hal
104	Gold	1499.00	12	Anr

- **FIND CUSTOMERS WHO SPENT MORE THAN AVERAGE**

SELECT * FROM CUSTOMERS WHERE CUSTOMER_ID IN(SELECT CUSTOMER_ID FROM SUBSCRIPTIONS);

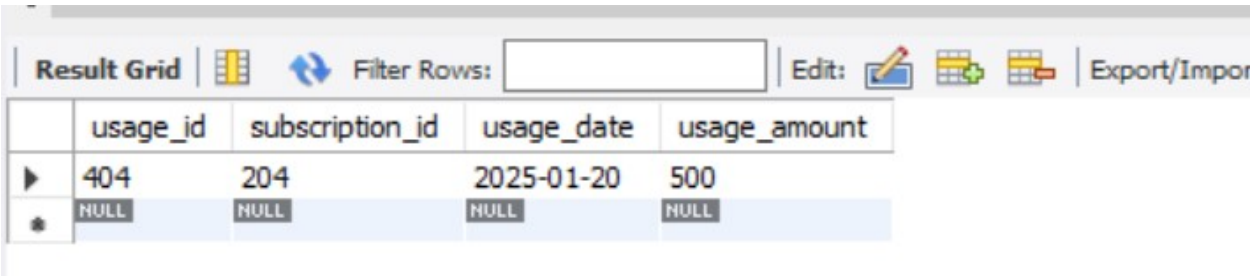


Result Grid | Filter Rows: | Edit: | Export/Impo

customer_id	name	email	phone	registration_date
1	Rahul Sharma	rahul@gmail.com	9876543210	2025-01-10
2	Priya Reddy	priya@gmail.com	9876543211	2025-02-15
3	Amit Kumar	amit@gmail.com	9876543212	2025-03-20

- **SHOW SUBSCRIPTIONS WITH HIGHEST USAGE**

SELECT * FROM USAGE_DETAILS WHERE USAGE_AMOUNT=(SELECT MAX(USAGE_AMOUNT) FROM USAGE_DETAILS);



Result Grid | Filter Rows: | Edit: | Export/Impor

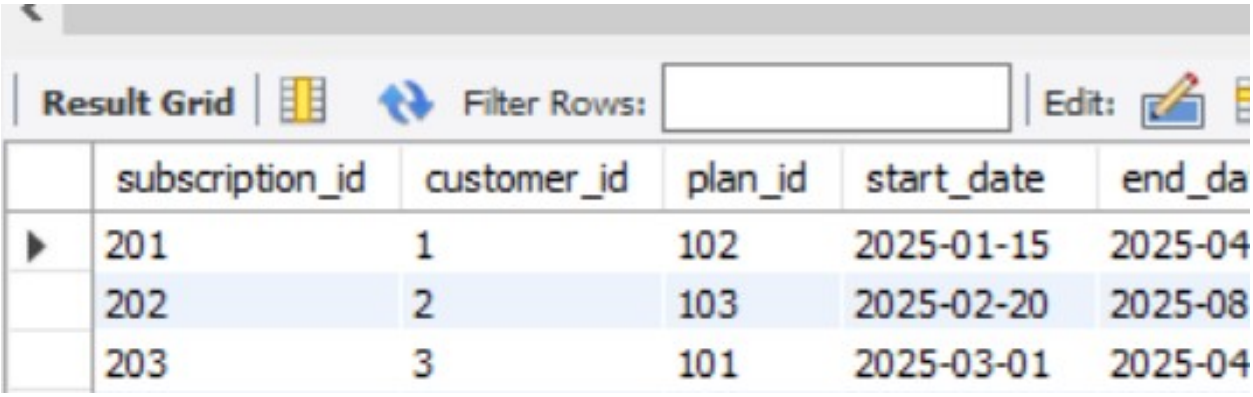
usage_id	subscription_id	usage_date	usage_amount
404	204	2025-01-20	500
NULL	NULL	NULL	NULL

UPDATES & DELETES:

- **UPDATE SUBSCRIPTION STATUS**

UPDATE SUBSCRIPTIONS SET STATUS='EXPIRED' WHERE SUBSCRIPTION_ID=201;

SELECT * FROM SUBSCRIPTIONS;



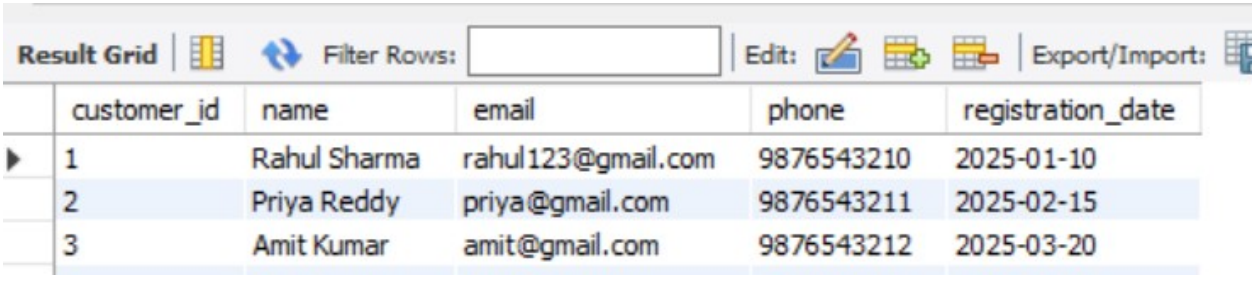
Result Grid | Filter Rows: | Edit:

subscription_id	customer_id	plan_id	start_date	end_da
201	1	102	2025-01-15	2025-04
202	2	103	2025-02-20	2025-08
203	3	101	2025-03-01	2025-04

- **CHANGE CUSTOMER EMAIL**

UPDATE CUSTOMERS SET EMAIL='rahul123@gmail.com' WHERE CUSTOMER_ID=1;

SELECT * FROM CUSTOMERS;



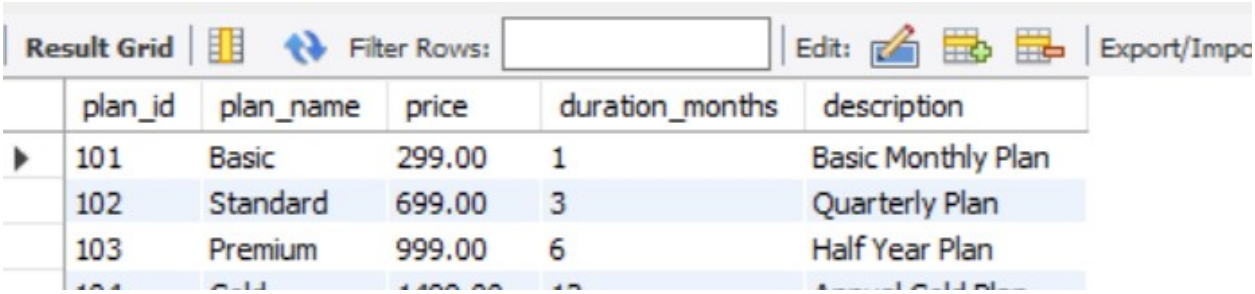
Result Grid | Filter Rows: | Edit: | Export/Import:

customer_id	name	email	phone	registration_date
1	Rahul Sharma	rahul123@gmail.com	9876543210	2025-01-10
2	Priya Reddy	priya@gmail.com	9876543211	2025-02-15
3	Amit Kumar	amit@gmail.com	9876543212	2025-03-20

- **UPDATE PLAN PRICE**

UPDATE PLANS SET PRICE=699 WHERE PLAN_ID=102;

SELECT * FROM PLANS;



plan_id	plan_name	price	duration_months	description
101	Basic	299.00	1	Basic Monthly Plan
102	Standard	699.00	3	Quarterly Plan
103	Premium	999.00	6	Half Year Plan

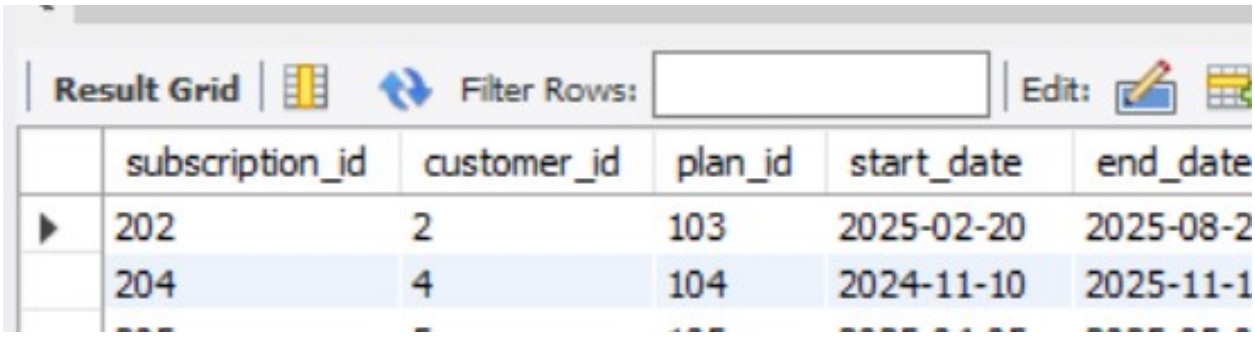
- **DEELETE EXPIRED SUBSCRIPTIONS**

DELETE FROM PAYMENTS WHERE SUBSCRIPTION_ID IN(201,203);

DELETE FROM USAGE_DETAILS WHERE SUBSCRIPTION_ID IN(201,203);

DELETE FROM SUBSCRIPTIONS WHERE STATUS='EXPIRED';

SELECT * FROM SUBSCRIPTIONS;

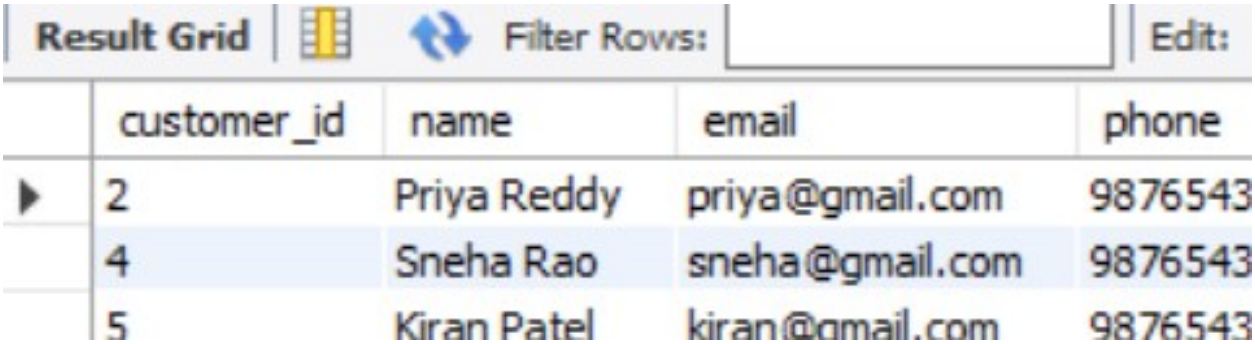


subscription_id	customer_id	plan_id	start_date	end_date
202	2	103	2025-02-20	2025-08-2
204	4	104	2024-11-10	2025-11-1

- **DELETE CUSTOMERS WITH NO SUBSCRIPTIONS**

DELETE FROM CUSTOMERS WHERE CUSTOMER_ID NOT IN (SELECT CUSTOMER_ID FROM SUBSCRIPTIONS);

SELECT * FROM CUSTOMERS;



customer_id	name	email	phone
2	Priya Reddy	priya@gmail.com	9876543
4	Sneha Rao	sneha@gmail.com	9876543
5	Kiran Patel	kiran@gmail.com	9876543

VIEWS

- **CREATE A VIEW TO DISPLAY CUSTOMER SUBSCRIPTIONS**

```
CREATE VIEW Customer_Subscriptions AS

SELECT c.name, p.plan_name, s.status

FROM Customers c

JOIN Subscriptions s

ON c.customer_id=s.customer_id

JOIN Plans p

ON s.plan_id=p.plan_id;
```

- **CREATE A VIEW FOR PAYMENT HISTORY**

```
CREATE VIEW Payment_History AS

SELECT *

FROM Payments;
```

- **CREATE A VIEW SHOWING ACTIVE SUBSCRIPTIONS**

```
CREATE VIEW Active_Subscriptions AS

SELECT *

FROM Subscriptions

WHERE status='Active';
```

- **CREATE A VIEW FOR PLAN DETAILS WITH DISCOUNTS**

```
CREATE VIEW Plan_Discounts AS

SELECT p.plan_name, p.price, d.discount_percentage

FROM Plans p

JOIN Discounts d

ON p.plan_id=d.plan_id;
```

- **CREATE A VIEW SHOWING CUSTOMER SPENDING**

```
CREATE VIEW Customer_Spending AS
```

```
SELECT c.name, SUM(p.amount) AS Total_Spending
FROM Customers c
JOIN Subscriptions s
ON c.customer_id=s.customer_id
JOIN Payments p
ON s.subscription_id=p.subscription_id
GROUP BY c.name;
```

STORED PROCEDURES:

1.INSERT NEW CUSTOMER

```
DELIMITER $$
CREATE PROCEDURE InsertCustomer(
IN p_name VARCHAR(100),
IN p_email VARCHAR(100),
IN p_phone VARCHAR(15),
IN p_date DATE
)
BEGIN
INSERT INTO Customers(customer_name,email,phone,registration_date)
VALUES(p_name,p_email,p_phone,p_date);
END $$
DELIMTER;
CALL InsertCustomer('Rahul','rahul@gmail.com','9876543210','2025-07-28');
```

2.CREATE SUBSCRIPTION

```
DELIMITER $$
```

```
CREATE PROCEDURE CreateSubscription(
IN p_customer_id INT,
IN p_plan_id INT,
IN p_start_date DATE
)
BEGIN
INSERT INTO Subscriptions
(subscription_id,customer_id,plan_id,start_date,end_date,status)
VALUES
(206,p_customer_id,p_plan_id,p_start_date,'2025-12-31','Active');
END $$
DELIMITER ;
CALL CreateSubscription(2,101,'2025-07-28');
```

3.GET CUSTOMER SUBSCRIPTIONS

```
DELIMITER $$
CREATE PROCEDURE GetCustomerSubscriptions(
IN p_customer_id INT
)
BEGIN
SELECT * FROM Subscriptions
WHERE customer_id=p_customer_id;
END $$
DELIMITER ;
CALL GetCustomerSubscriptions(2);
```

4.CALCULATE TOTAL REVENUE

```
DELIMITER $$

CREATE PROCEDURE TotalRevenue()

BEGIN

SELECT SUM(amount) AS Total_Revenue

FROM Payments;

END $$

DELIMITER ;
```

5.LIST PLANS BY PRICE RANGE

```
DELIMITER $$

CREATE PROCEDURE PlanPriceRange(

IN min_price DECIMAL(10,2),

IN max_price DECIMAL(10,2)

)

BEGIN

SELECT * FROM Plans

WHERE price BETWEEN min_price AND max_price;

END $$

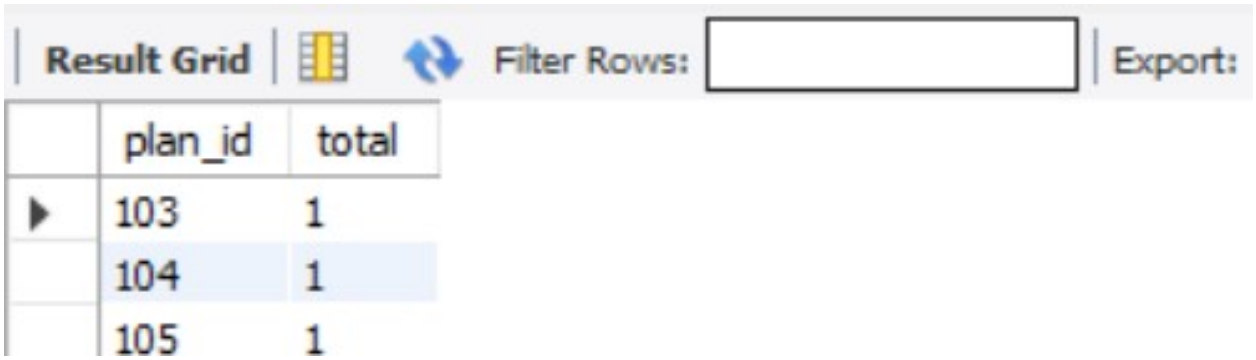
DELIMITER ;

CALL PlanPriceRange(500,1000);
```

REPORTS/ANALYSIS

1.TOP 5 MOST POPULAR PLANS

```
SELECT plan_id,count(*) as total from subscriptions group by plan_id order by total desc limit 5;
```



plan_id	total
103	1
104	1
105	1

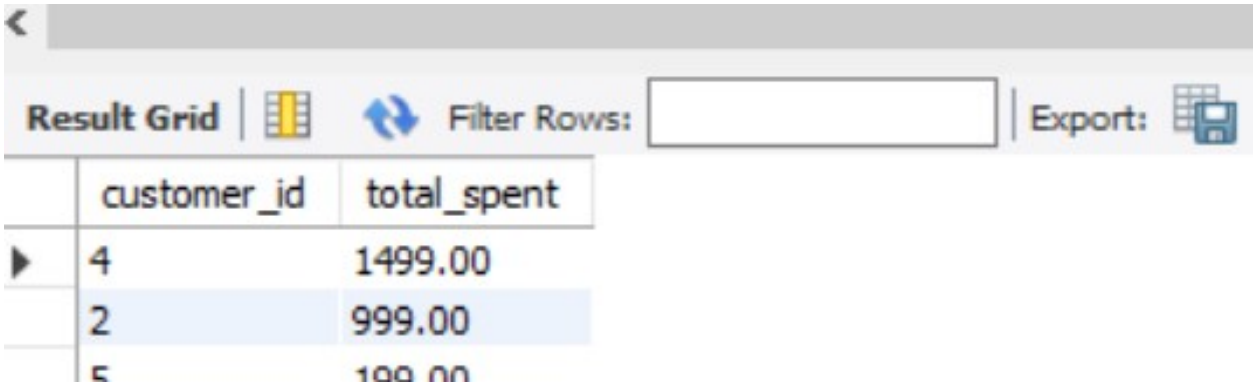
2.CUSTOMERS WITH HIGHEST SPENDING.

SELECT customer_id,sum(p.amount) as total_spent from payments p join subscriptions s on p.subscription_id = s.subscription_id

group by s.customer_id

order by total_spent desc

limit 5;



customer_id	total_spent
4	1499.00
2	999.00
5	199.00

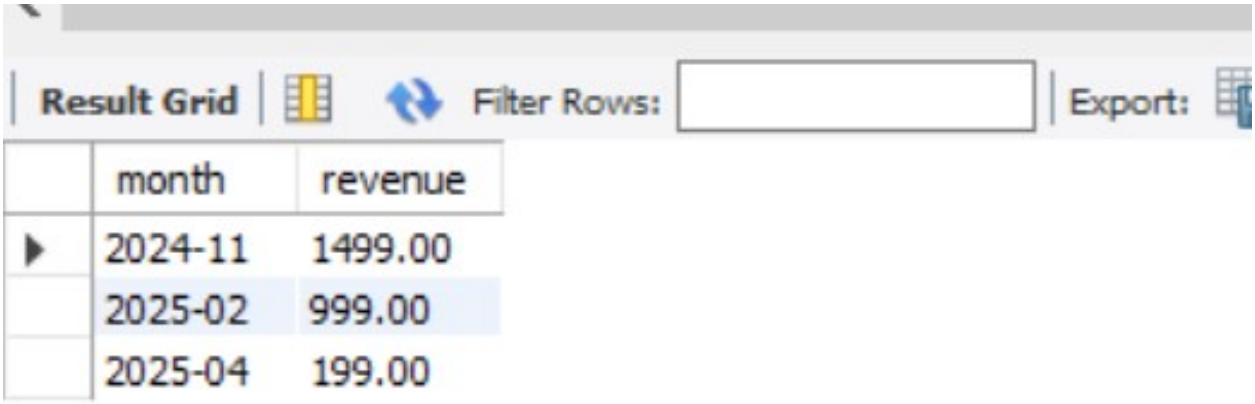
3.MONTHLY REVENUE REPORT

SELECT DATE_FORMAT(payment_date, '%Y-%m') AS month, SUM(amount) AS revenue

FROM payments

GROUP BY month

ORDER BY month;



	month	revenue
▶	2024-11	1499.00
	2025-02	999.00
	2025-04	199.00

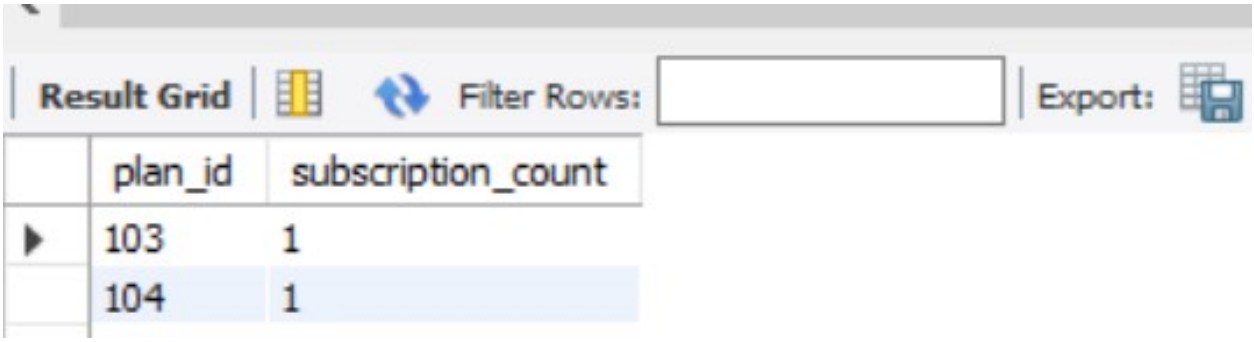
4.PLAN-WISE SUBSCRIPTION COUNT

```
SELECT plan_id, COUNT(*) AS subscription_count

FROM subscriptions

GROUP BY plan_id

ORDER BY subscription_count DESC;
```



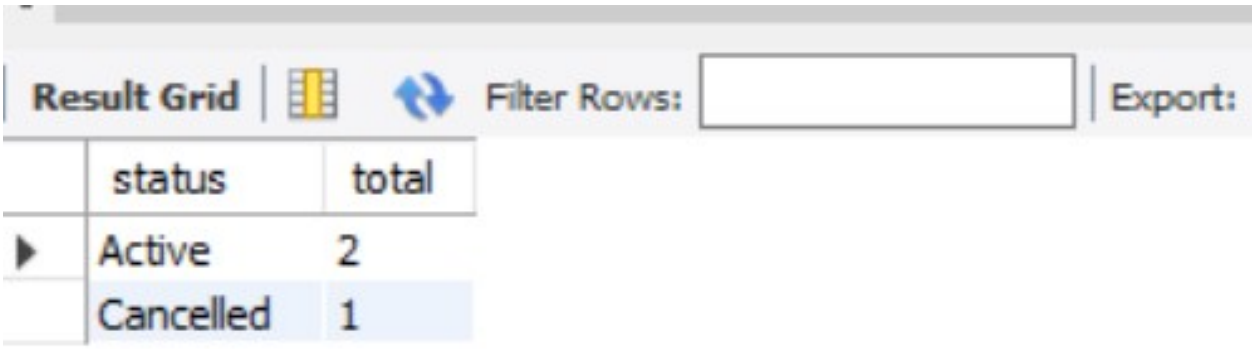
	plan_id	subscription_count
▶	103	1
	104	1

5.SUBSCRIPTION CHURN ANALYSIS(ACTIVE VS CANCELLED)

```
SELECT status, COUNT(*) AS total

FROM subscriptions

GROUP BY status;
```



status	total
Active	2
Cancelled	1